Abstract

Title: An Independence Platform for Voice-Native OS Navigation by Individuals with Severe Motor Impairment

Fully paralyzed individuals who retain speech have no adequate way to independently operate a computer — existing voice tools are either rigid command-based systems that can't handle complex intent, or UI agents confined to browser automation. We propose a voice-driven navigation agent operating at the full OS level, using an LLM to translate natural, unstructured speech into executable actions via real-time code generation, enabling complex multi-step tasks beyond what command-based systems can parse. To close the perception-action loop, the agent captures and interprets screen state — via vision-model screenshot analysis or a lighter accessibility-tree read — before and after each action, verifying that clicks, field entries, or dialogs behaved as expected and self-correcting when they don't, since the user cannot physically intervene if something misfires.

For real-time responsiveness, we use models with native audio understanding rather than a cascaded STT→LLM→TTS pipeline, avoiding compounded latency at each stage. We integrate domestic language models such as Sarvam AI for regional Indian language support, removing the English-only barrier common in existing assistants. The core challenges are minimizing end-to-end inference latency across the audio, vision-verification, and code-generation stages — critical for a tool replacing physical interaction in real time — and preserving that responsiveness in a fully offline configuration, where local model constraints typically trade speed and accuracy for independence from connectivity.

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